

A Data Collection

Fragmented public skill-security resources



13
Public
Sources



11
Core-
contributing
Sources

Heterogeneity across dimensions



Artifact
Form



Provenance



Label
Evidence



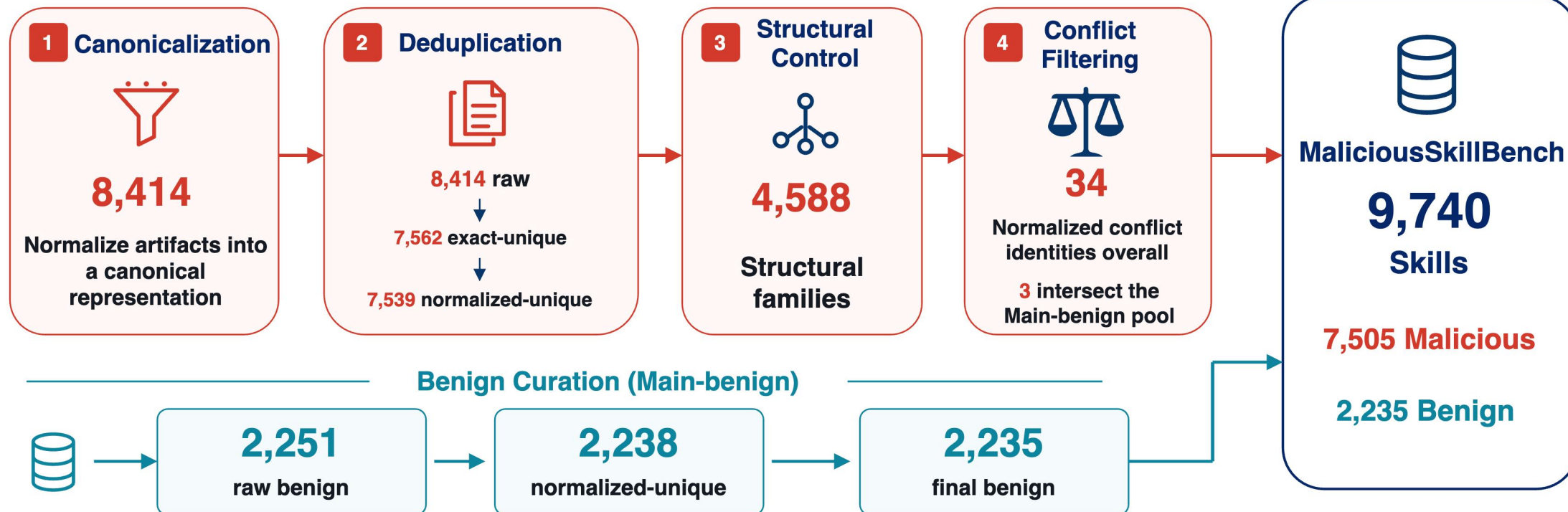
Benign
Support

*Heterogeneous formats, cross-source reuse,
and uneven benign support*

B Benchmark Construction

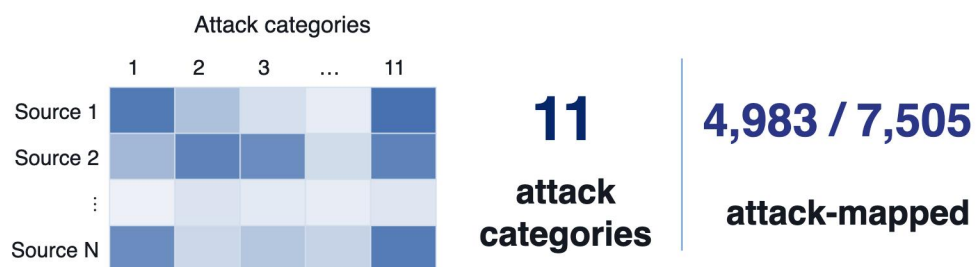
Consolidate, not concatenate.

Malicious Construction (Core)



C Characterization & Evaluation

1 Threat Diversity



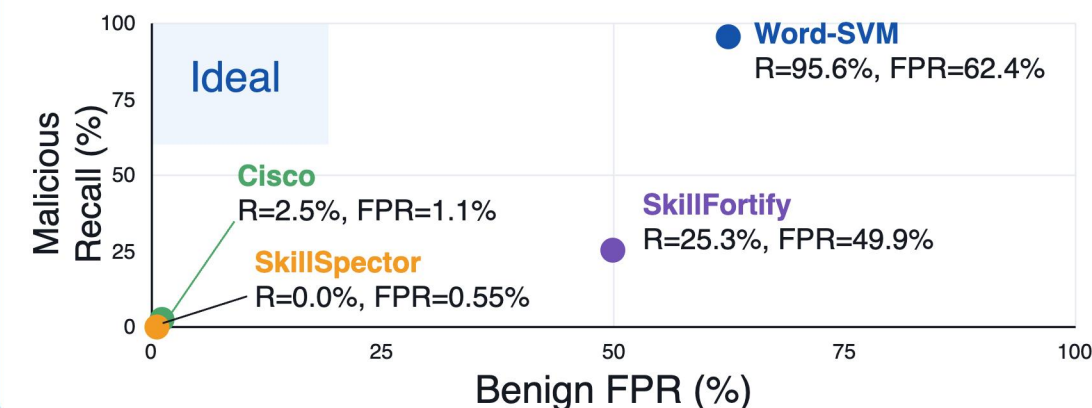
Distinct sources exhibit different threat profiles.

2 Distribution Shift



Largest drop under source shift.

3 Detector Trade-off



★ **Detection Gap:** No evaluated method jointly achieves high malicious recall and low benign false-positive rates under source shift.